PRP PROJECT

E-Commerce Digital Analytics

Final Deliverable Presentation

SQL Server • Python & Machine Learning • Power BI

End-to-end analytics on website sessions, orders, products & refunds

DATABASE

PRP_Ecommerce_Analytics

ANALYSIS WINDOW

Dec 2012 – Apr 2015

SUBMISSION

2nd September 2026

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AGENDA

What We'll Cover Today

01 **Business Understanding**

Context, problem statement & objectives

02 **Technology Stack & Data Architecture**

Tools used and how the data is modelled

03 **Data Quality & KPI Framework**

Audit approach and how KPIs are defined

04 **Descriptive Analysis**

Revenue, marketing, product & refund performance

05 **Diagnostic Analysis**

Where and why performance varies

06 **Predictive Analysis**

Session conversion model & key drivers

07 **Power BI Dashboards**

Executive-ready visual reporting

08 **Findings, Recommendations & Conclusion**

What we learned and what to do next

BUSINESS UNDERSTANDING

Business Context & Problem Statement

Context

PRP operates an online store selling a small catalogue of gift and novelty products (e.g. “The Original Mr. Fuzzy”). The business tracks every website session, order, product line item and refund, but this data has never been consolidated into a single source of truth or turned into a decision-ready reporting layer.

Problem Statement

- Leadership cannot see revenue, margin and refund performance in one place.
- It is unclear which marketing channels and devices actually convert profitably.
- Refunds and margin erosion by product are not systematically investigated.
- There is no data-driven way to flag sessions likely (or unlikely) to convert.

What Success Looks Like

- ➤ One clean SQL Server model behind every number
- ➤ Self-serve Power BI dashboards for execs & marketing
- ➤ A documented, repeatable EDA + diagnostic process
- ➤ A predictive model that flags conversion likelihood
- ➤ Clear, prioritised business recommendations

OBJECTIVES & TECH STACK

Project Objectives & Technology Stack

Objectives

1. Build a validated, single source of truth in SQL Server
2. Deliver descriptive & diagnostic analysis of sales, marketing, product and refund performance
3. Design and build Power BI dashboards for ongoing self-serve reporting
4. Develop a predictive (logistic regression) model for session conversion
5. Translate analysis into prioritised, actionable business recommendations

Technology Stack

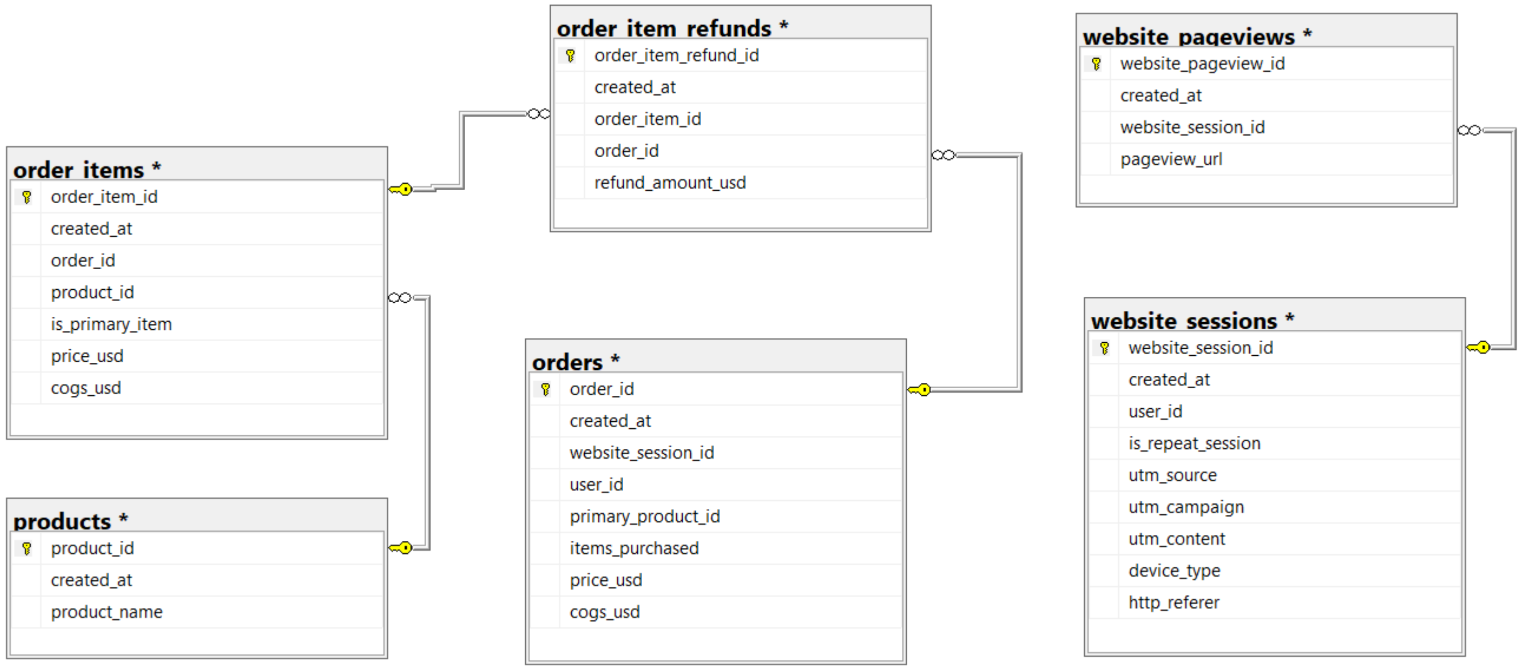
1. **SQL Server**
   Data warehouse, cleaning views, KPI & diagnostic queries
2. **Python (pandas, scikit-learn)**
   EDA, visualisation & logistic regression modelling
3. **Jupyter Notebook**
   Reproducible, documented analysis workflow
4. **Power BI Desktop**
   Executive, marketing, product & refund dashboards
5. **Project Documentation**
   Structured documentation of SQL, Python & Power BI analysis

Every objective above is delivered through this exact stack – SQL Server for the source of truth (connected via pyodbc), Python for analysis and modelling, Power BI for the reporting layer, and GitHub to keep it all versioned.

DATA ARCHITECTURE

Data Model & ER Diagram

6 core tables covering sessions, pageviews, orders, order items, products & refunds



Source: PRP_Ecommerce_Analytics – dbo schema (raw) → analytics schema (clean views)

Core Tables

- website_sessions – traffic source, device, repeat flag
- website_pageviews – page-level browsing path
- products – product catalogue
- orders – order-level revenue & COGS
- order_items – line-item detail per order
- order_item_refunds – refund amount & timing

DATA QUALITY

Data Audit & Cleaning Approach

10 SQL scripts, executed in order, transform raw tables into trusted analytics views

1. **Audit Revalidation**

   Row counts, primary-key duplicate checks and meaningful-NULL review on all 6 source tables.

2. **Cleaning & Transformation**

   Standardised types, trimmed text fields and built analytics.vw_* clean views without altering raw tables.

3. **KPI Definitions**

   Reusable queries for revenue, margin, AOV, refunds, sessions & conversion rate.

4. **EDA (Sales, Marketing, Product, Customer)**

   Four dedicated scripts profiling each business area independently.

5. **Diagnostic Analysis**

   Comparative & decomposition queries – device, channel, product and calendar cuts.

6. **Final Reconciliation & KPI Snapshot**

   Single validated result set used as the source of truth for this presentation.

Principle

Raw source tables are never modified. Every transformation lives in versioned SQL views (analytics schema), so results are always reproducible and auditable back to source.

KPI FRAMEWORK

How Our KPIs Are Defined

Definitions used consistently across SQL, Python and Power BI

KPI	Definition	Layer
Gross Revenue	SUM(price_usd) across all order line items	SQL
Gross Margin %	(Revenue − COGS) / Revenue × 100	SQL
Average Order Value (AOV)	AVG(price_usd) per order	SQL
Net Revenue	Gross Revenue − Total Refund Amount	SQL
Conversion Rate	Converted Sessions / Total Sessions × 100	SQL / Python
Revenue per Session	Gross Revenue / Total Sessions	SQL / Power BI
Refund Rate	Refund Amount / Revenue × 100	SQL / Power BI

DESCRIPTIVE ANALYSIS

Executive KPI Snapshot

SQL final KPI snapshot – Dec 2012 to Apr 2015, validated against Power BI

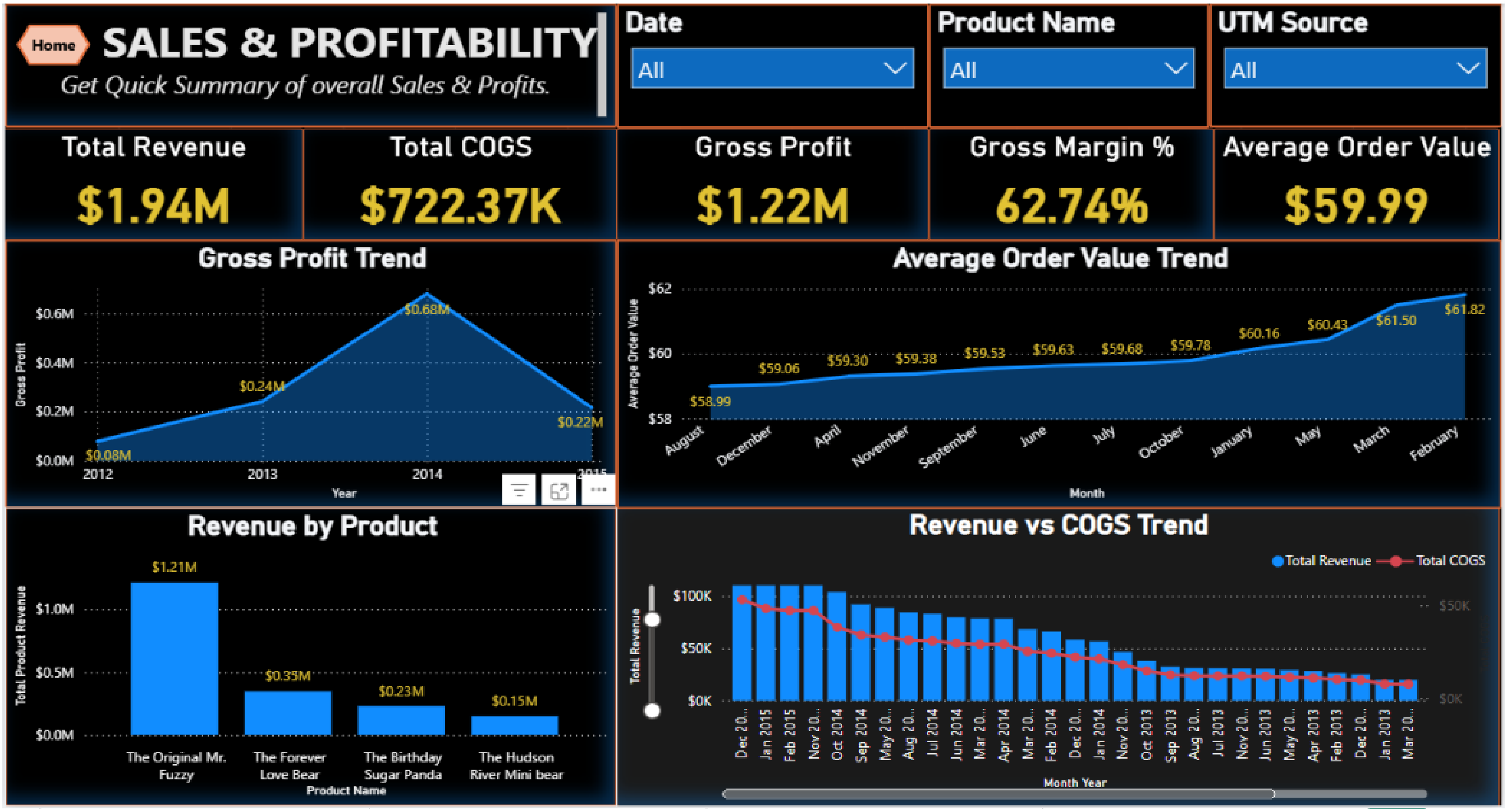
$1.94M Gross Revenue	**32,313** Total Orders	**472,871** Website Sessions	**6.83%** Conversion Rate
$59.99 Avg. Order Value	**62.74%** Gross Margin	**$1.22M** Gross Profit	**$85.3K** Total Refunds (4.40% of rev.)

Net Revenue: $1.85M • Total COGS: $722.4K • Revenue per Session: $4.10

DESCRIPTIVE ANALYSIS

Revenue & Profitability Trends

Power BI – Sales & Profitability dashboard, connected live to SQL Server



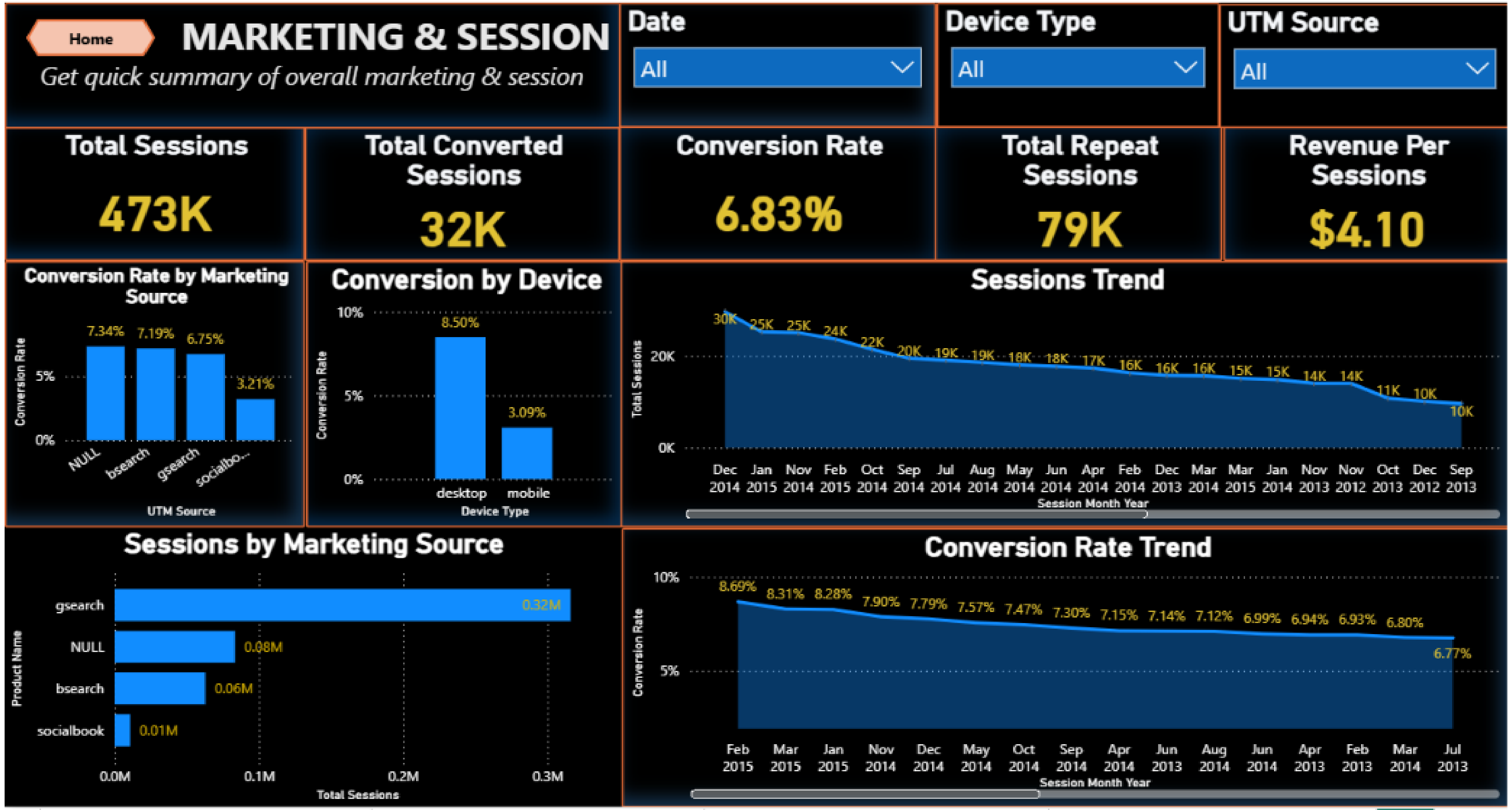
Reading the Trend

- Revenue & gross profit peak in 2014, then taper into early 2015
- AOV climbs steadily from $58.99 to $61.82 over the window
- COGS tracks revenue closely, keeping margin stable near 62–63%
- Order volume declines faster than revenue – fewer, higher-value orders

DESCRIPTIVE ANALYSIS

Marketing & Session Performance

Power BI – Marketing & Sessions dashboard



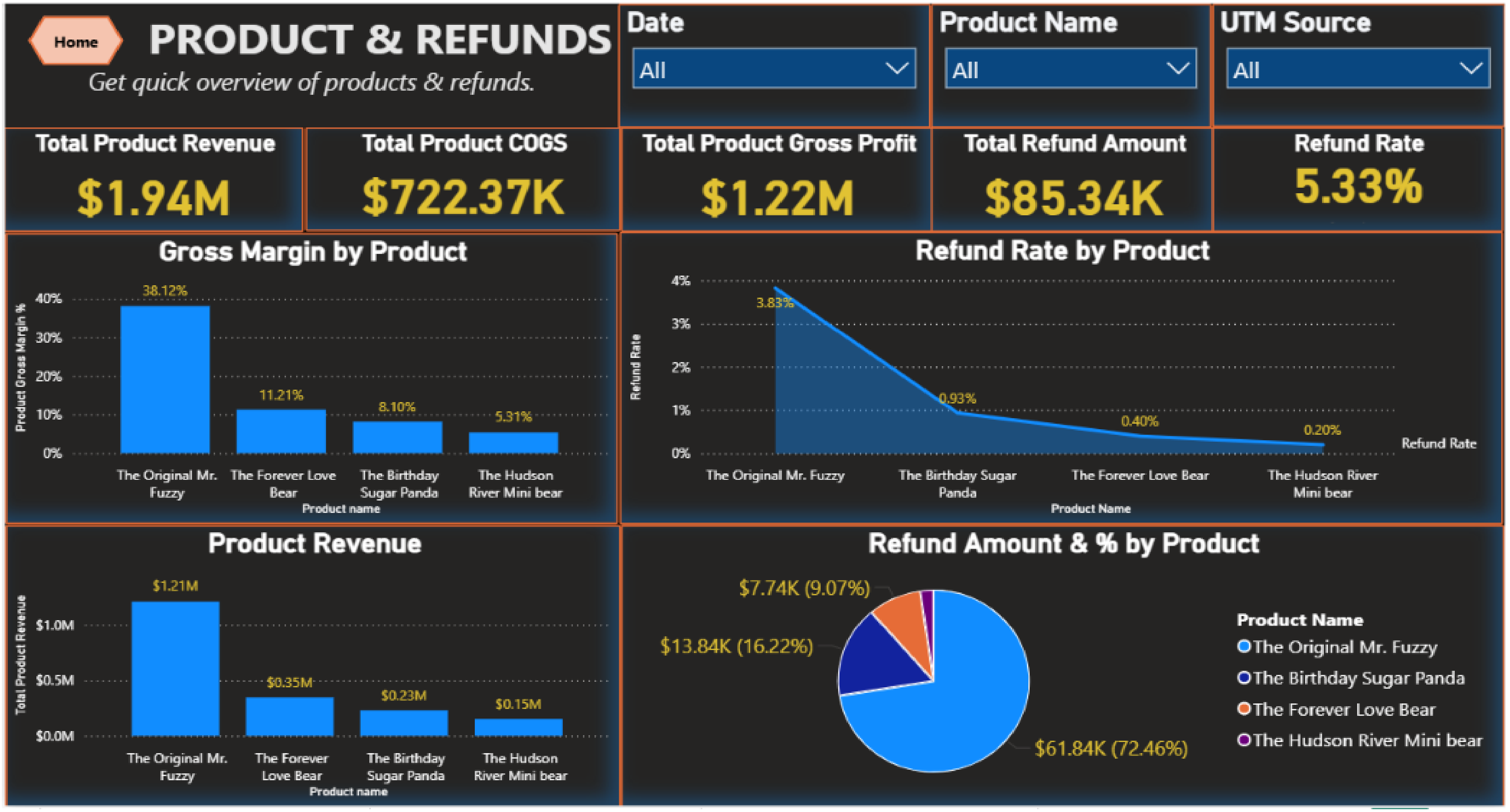
Reading the Trend

- Google Search drives 67% of all sessions (316K of 473K)
- Desktop converts nearly 3x better than mobile (8.50% vs 3.09%)
- Repeat sessions are only 17% of traffic but convert better (7.83% vs 6.64%)
- Overall conversion rate has drifted down from ~8.7% to ~6.8%

DESCRIPTIVE ANALYSIS

Product Performance & Refunds

Power BI – Product & Refunds dashboard



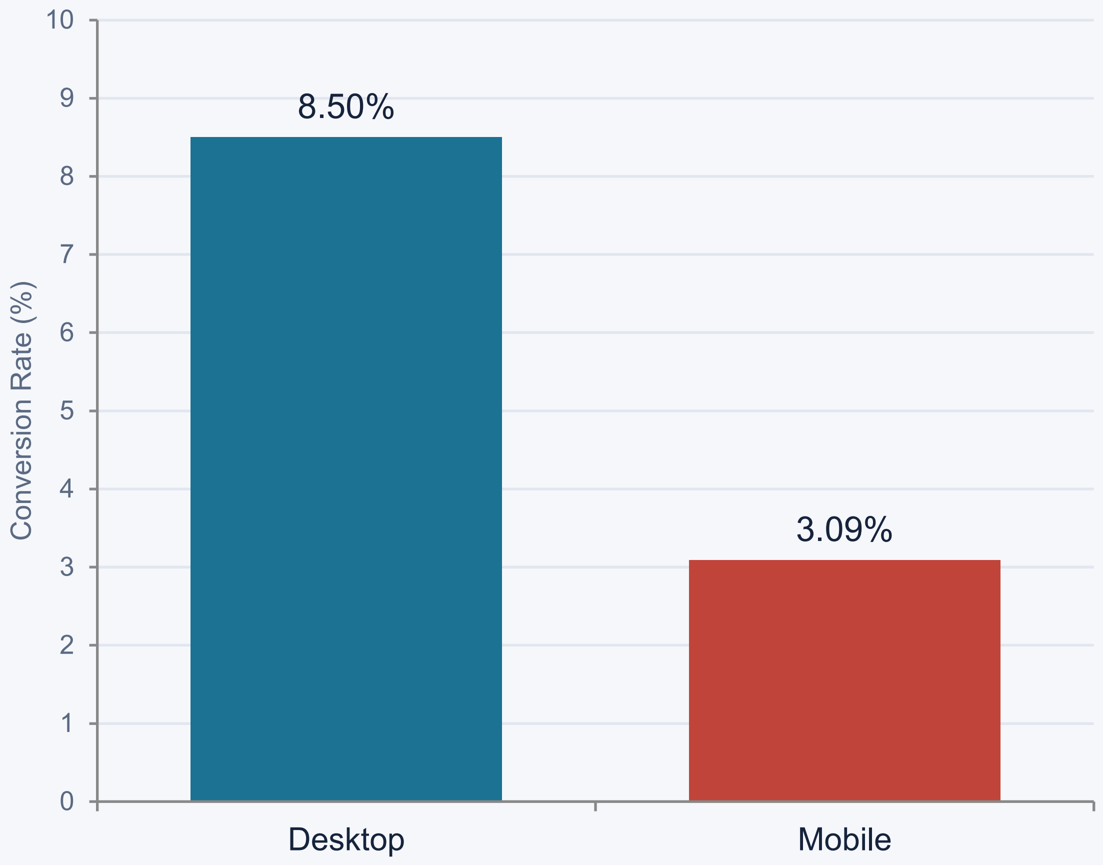
Reading the Trend

- Mr. Fuzzy drives 62% of revenue but has the thinnest margin (61.0%)
- Birthday Sugar Panda has the best margin (68.5%) and highest refund rate (6.04%)
- Hudson River Mini Bear is the safest line: strong margin, lowest refunds (1.28%)
- 72% of all refund dollars come from Mr. Fuzzy alone ($61.8K)

DIAGNOSTIC ANALYSIS

Where Does Conversion Fall Short?

Conversion rate by device — the single biggest performance gap in the funnel



Diagnostic Read-outs

Device gap

Mobile converts at 3.09% vs desktop 8.50% – a 5.4 point gap despite mobile carrying 31% of all sessions (145.8K).

Session type

Repeat sessions convert better than new (7.83% vs 6.64%), but make up only 17% of traffic – upside in remarketing.

Revenue at risk

At desktop's conversion rate, mobile's 145.8K sessions would generate roughly $740K – about $470K more than today.

DIAGNOSTIC ANALYSIS

Channel Quality & Product Risk

Volume is not the same as value — looking beyond session/unit counts

Marketing Channel Quality Matrix

Channel	Sessions	Conv. Rate	Rev / Session
Direct / Organic (NULL)	83,328	7.34%	$4.46
bsearch	62,823	7.19%	$4.28
gsearch	316,035	6.75%	$4.04
socialbook	10,685	3.21%	$2.08

gsearch drives the most sessions and revenue, but direct/organic and bsearch traffic actually convert more efficiently per session.

Product Margin vs. Refund Risk

Product	Margin %	Refund %
The Original Mr. Fuzzy	61.0%	5.11%
The Forever Love Bear	62.5%	2.23%
The Birthday Sugar Panda	68.5%	6.04%
The Hudson River Mini Bear	68.4%	1.28%

So What?

Rebalance channel spend

socialbook converts at less than half the average rate and returns the lowest revenue per session — review targeting and creative before increasing spend.

Best-margin product carries the most refund risk

Birthday Sugar Panda pairs the highest margin (68.5%) with the highest refund rate (6.04%) – a quality or expectation-setting review would protect that margin.

Flagship product needs a refund review

Mr. Fuzzy is 62% of revenue and 72% of refund dollars – even a small improvement here has an outsized profit impact.

PREDICTIVE ANALYSIS

Predicting Session Conversion

Approach, feature engineering & model selection

Modelling Approach

Objective

Predict whether a website session converts into an order (binary classification).

Target

converted_session (1 = converted, 0 = not converted)

Algorithm

Logistic Regression – chosen for a binary target and for interpretable, business-explainable coefficients.

Leakage control

order_count and session_revenue intentionally excluded – they reveal the outcome directly.

Features used

Device type, UTM source/campaign/content, referrer, repeat-session flag, session hour, day of week, month & year.

Why Logistic Regression Fits Here

- Target is binary (converted vs not converted) – a natural fit for logistic, not linear, regression
- Coefficients translate directly into “which session traits raise or lower conversion odds” – useful for marketing conversations
- Fast to train and re-train as new session data lands, keeping the model easy to refresh
- Serves as an interpretable baseline before considering tree-based models for further lift

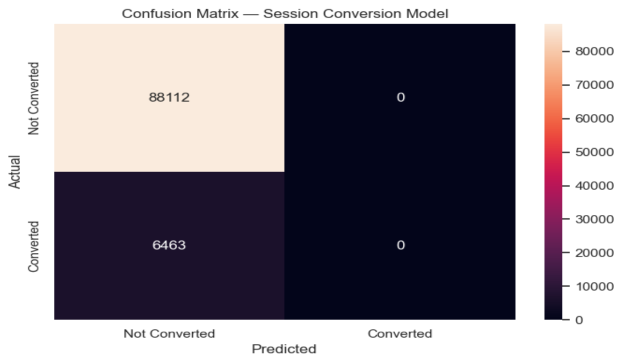
Pipeline

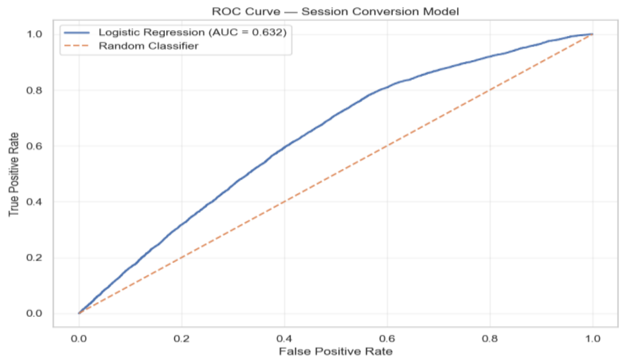
SQL Server → Python extraction → validation → train/test split → preprocessing (encode + scale) → Logistic Regression → evaluation → model saved for reuse

PREDICTIVE ANALYSIS

Model Results & Key Drivers

Held-out test set of 94,575 sessions





93.17%
Accuracy

0.632
ROC-AUC

0.00
Precision (class 1)

Honest Read of Model Quality

The dataset is imbalanced – only about 7% of sessions convert. The model reaches 93% accuracy simply by leaning toward the majority "not converted" class, so precision/recall on actual conversions are effectively zero at the default threshold. ROC-AUC of 0.63 shows the model has real, moderate signal – useful for ranking session quality – but it is not yet reliable as a stand-alone converter/non-converter classifier. Next step: address class imbalance (class weighting / resampling) and tune the decision threshold before deployment.

What Drives Conversion Odds

Raises odds:

Later years/months in the dataset • Repeat session • Time-of-day / weekday effects (small)

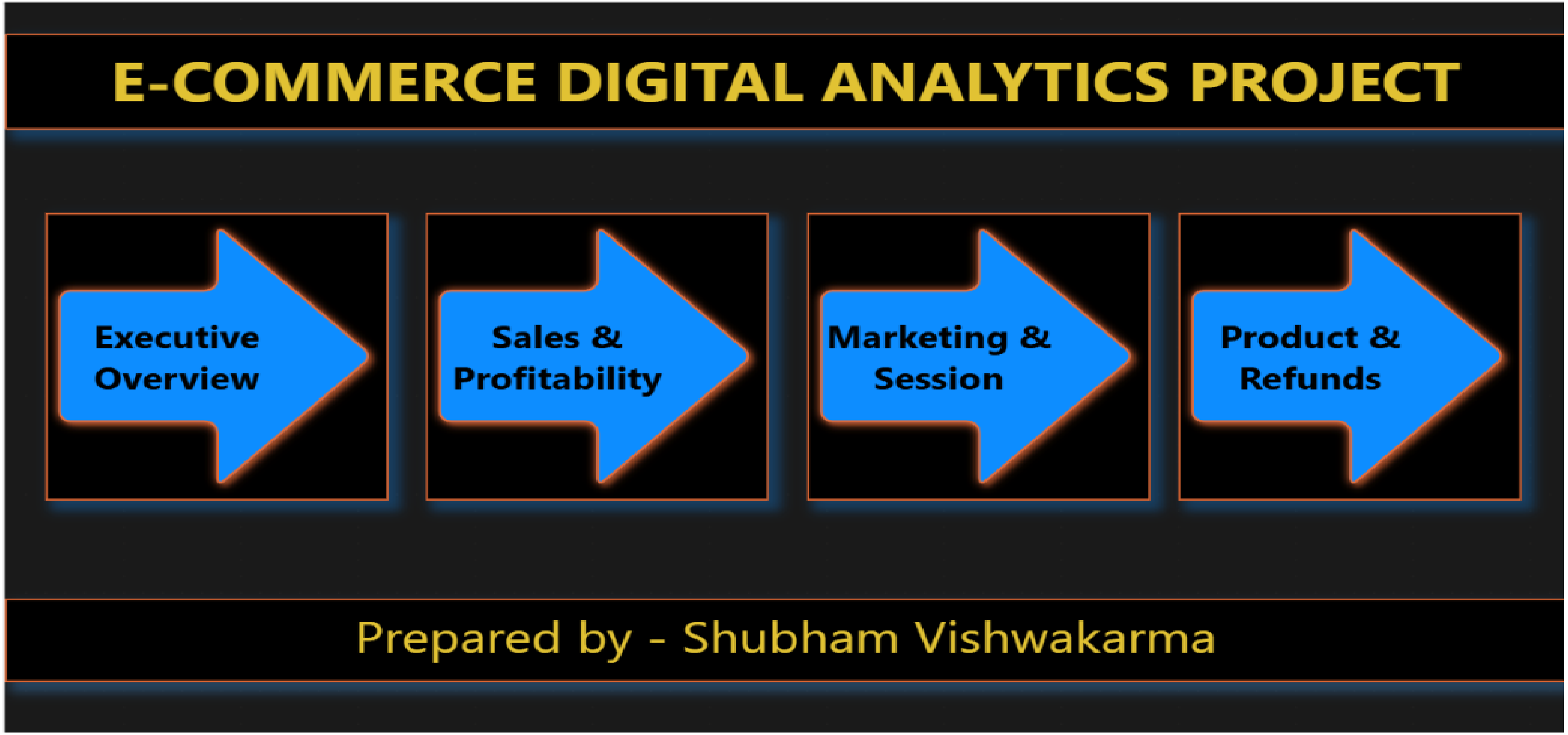
Lowers odds:

Mobile device (strongest negative driver) • Social (socialbook) traffic • Paid non-brand / pilot campaigns • bsearch source

Confirms the diagnostic finding: mobile and social traffic are the clearest places to improve conversion.

Power BI Reporting Hub

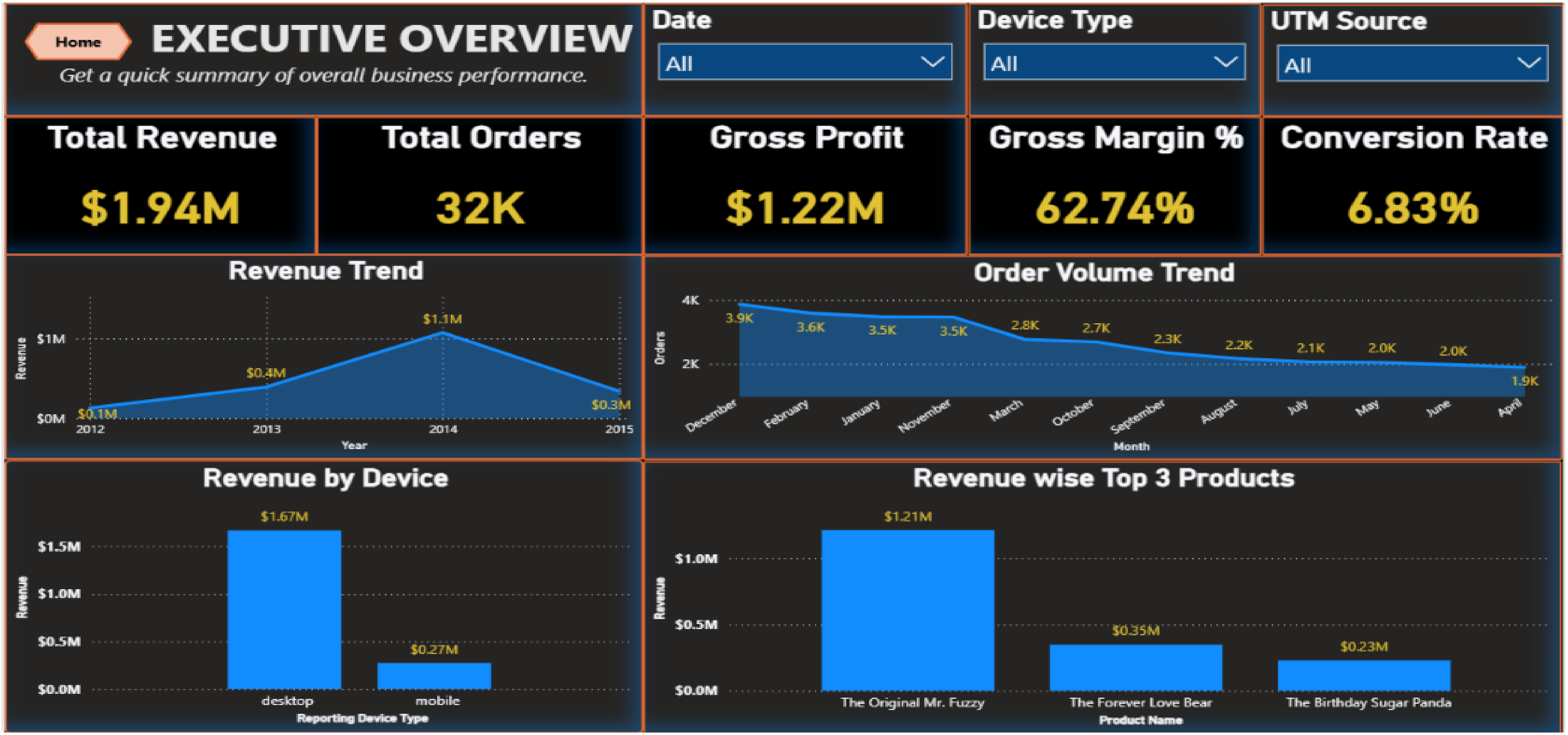
Four published dashboards, one navigation home page – built and shared by Shubham Vishwakarma



POWER BI REPORTING

Executive Overview Dashboard

Dynamic, filterable by date, device type & UTM source – published to Power BI Service



FINDINGS

Key Business Findings

Synthesised from SQL, Python and Power BI analysis

Revenue is concentrated

The Original Mr. Fuzzy alone drives 62% of revenue and 61% of gross profit – the business is highly dependent on one SKU.

Mobile is under-converting

Mobile sessions convert at 3.09% vs 8.50% on desktop — the single largest, most fixable performance gap identified.

Paid search scales volume, not efficiency

gsearch brings the most sessions and revenue, but direct/organic and bsearch traffic convert more efficiently per session.

Margin and refund risk can co-exist

The Birthday Sugar Panda has the best margin (68.5%) and the highest refund rate (6.04%) – worth a quality/expectations review.

Repeat visitors are more valuable

Repeat sessions convert at 7.83% vs 6.64% for new sessions and generate $0.77 more revenue per session.

The conversion model shows moderate, usable signal

ROC-AUC of 0.63 confirms device and channel are real predictors, but class imbalance limits it as a stand-alone classifier today.

RECOMMENDATIONS

Business Recommendations

Prioritised, actionable next steps for the business

1

Fix the mobile experience

Run a UX/checkout audit on mobile; closing even half the gap to desktop could add roughly $200–250K in annual revenue.

2

Rebalance marketing spend on efficiency, not volume

Shift incremental budget toward channels with strong revenue-per-session (direct/organic, bsearch) and re-evaluate socialbook creative/targeting.

3

Protect the flagship product's margin & quality

Investigate root causes of Mr. Fuzzy's refund volume (72% of all refund dollars) – quality control, sizing/description accuracy, or packaging.

4

Review high-margin, high-refund products

Audit Birthday Sugar Panda for expectation-setting or quality issues before scaling marketing spend behind its strong margin.

5

Invest in retention & remarketing

Repeat sessions convert better and are still a small share of traffic – email/remarketing to past visitors is a high-ROI lever.

6

Mature the predictive model before deployment

Address class imbalance and tune the decision threshold; use it first as a session-scoring / prioritisation tool, not a hard gate.

CONCLUSION

Conclusion & Learnings

What This Project Delivered

- An end-to-end pipeline: SQL Server → Python → Power BI, built on validated, reproducible clean views
- Descriptive & diagnostic analysis covering sales, marketing, device, product and refund performance
- A working, interpretable logistic regression model for session conversion, with honest evaluation
- Prioritised, data-backed recommendations leadership can act on immediately

Learnings & Next Steps

- Clean, well-documented SQL views made every downstream tool (Python, Power BI) trustworthy and fast to build on
- Descriptive dashboards surfaced the mobile conversion gap before any modelling was needed – diagnostics matter as much as ML
- Class imbalance is a real constraint on this dataset; the next iteration should test class weighting, resampling and alternative models
- Next steps: monitor conversion performance, improve model evaluation, and re-run this analysis on newer data periodically

PRP PROJECT

Thank You